

Simulation-Based Investigation of Vaccination and Herd Immunity

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1. Introduction

Mathematical and computational epidemic models provide a controlled way to study how infectious diseases propagate through a population. The classical SIR model divides a population into susceptible, infectious and recovered groups. In this project, vaccination introduces a fourth state, producing an SIRV-style simulation in which vaccinated persons are immune from the beginning of a run. A central concept investigated in the simulation is herd immunity. When a sufficiently large fraction of the population is immune, an infectious person is less likely to encounter a susceptible person and sustained disease transmission becomes increasingly difficult. The purpose of this experiment was to examine this effect numerically by varying the vaccination rate from 0% to 100% and observing disease duration, the number of recovered persons, and the vaccinated population. The disease parameters assigned for this work were an infectious duration of 7 days and a transmissibility of $b = 0.3$. The simulation population contained 30,000 persons. The generated results were analysed using summary plots and a representative disease-progression curve.

2. Methodology

The C++ simulation explicitly tracks the state of each person. During every daily timestep, infectious persons make random contacts and susceptible contacts may become infected according to the transmissibility probability. An infected person remains infectious for seven days before entering the recovered state. Vaccinated persons remain immune and therefore do not participate in the susceptible-to-infectious transition. Vaccination rates were sampled from 0% to 100% in increments of 10%. For each vaccination level, the simulation records population-state statistics and the time until no infectious persons remain. The output was then used to compare the epidemic response across vaccination levels.

Parameter	Symbol	Value	Description
Disease duration	D	7 days	Time a person remains infectious
Transmissibility	β	0.3	Infection probability per contact
Contacts per day	C	6	Random contacts per infectious person
Population size	N	30000	Persons in the simulated population
Vaccination sweep	-	0-100%	Incremented in steps of 10 %

Table 1: Simulation Parameters

3. Results

Figure 1 shows the temporal progression of the four population states for a representative run at 30% vaccination. The vaccinated population remains constant at approximately 9,000 persons. The susceptible population initially contains about 21,000 persons and decreases rapidly once transmission accelerates. The infectious population rises sharply and reaches its maximum around day 17, at roughly 18,500 persons, before declining towards zero. At the same time, the recovered population increases and finally levels off close to 21,000 persons.

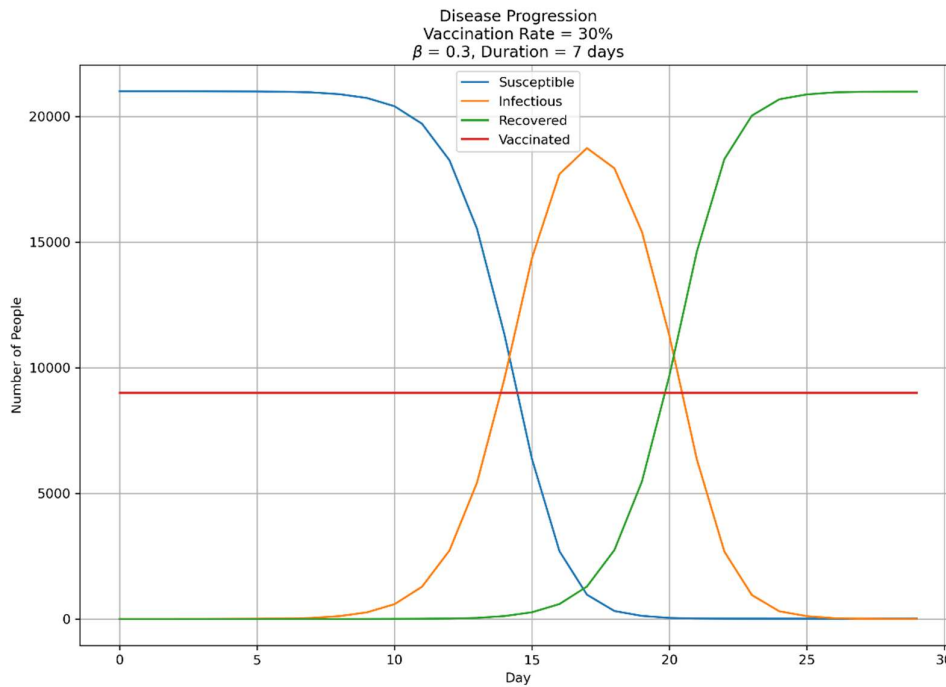


Figure 1: Disease progression at 30% vaccination ($\beta = 0.3$, $D = 7$)

The curve demonstrates the characteristic epidemic pattern produced by the simulation. At first, the number of infections is low. Once sufficient infectious persons are present, the epidemic enters a rapid growth phase. The infectious peak is followed by a decline because the susceptible pool has been strongly depleted and an increasing fraction of the population has recovered and become immune. The relatively large infectious peak is consistent with the seven-day disease duration: infected persons remain capable of transmitting the disease for several consecutive timesteps. Despite 30% vaccination, a substantial susceptible population remains available, so vaccination at this level is insufficient to prevent a large outbreak.

Figure 2 compares vaccination rate with the average number of days until disease extinction. The extinction time increases from approximately 24 days without vaccination to a maximum of about 71 days at 80% vaccination. It then drops sharply at 90% and reaches zero at 100% vaccination.

The non-monotonic behaviour of extinction time is important. A longer epidemic does not necessarily indicate a larger outbreak. Near the suppression threshold, transmission can

continue through small stochastic chains for many days even though the total number of infected persons is much lower. This provides a plausible explanation for the long average duration around 80% vaccination and the subsequent collapse at higher vaccination levels.

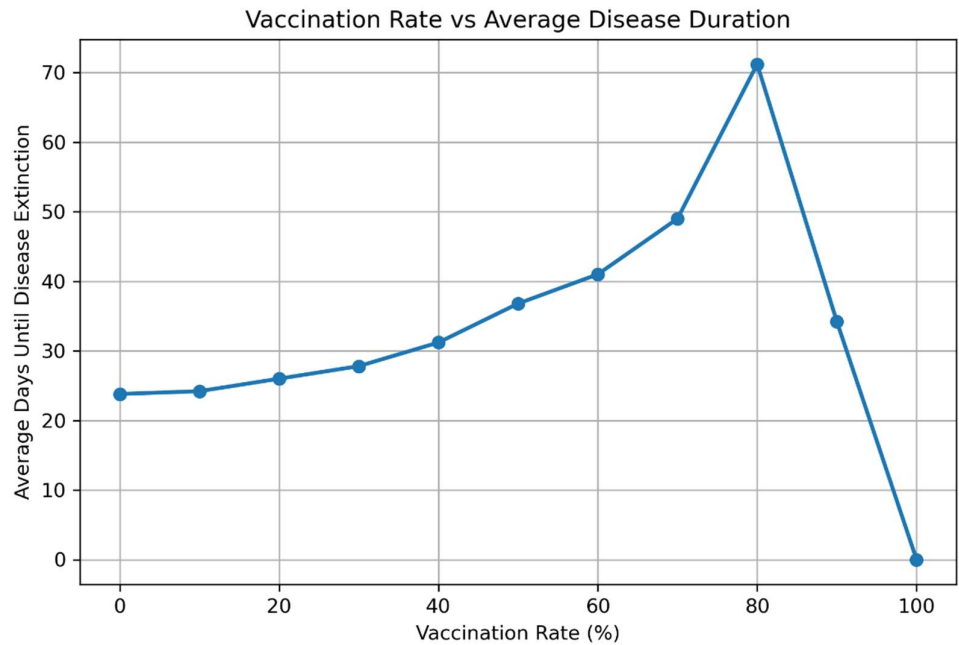


Figure 2: Vaccination rate versus average time until disease extinction

Figure 3 shows a strong decrease in the average recovered population as vaccination increases. Nearly the entire population is recovered at 0% vaccination, whereas the recovered count approaches zero around 90–100% vaccination. Because recovery follows infection in the model, this curve directly indicates that higher vaccination coverage prevents infections.

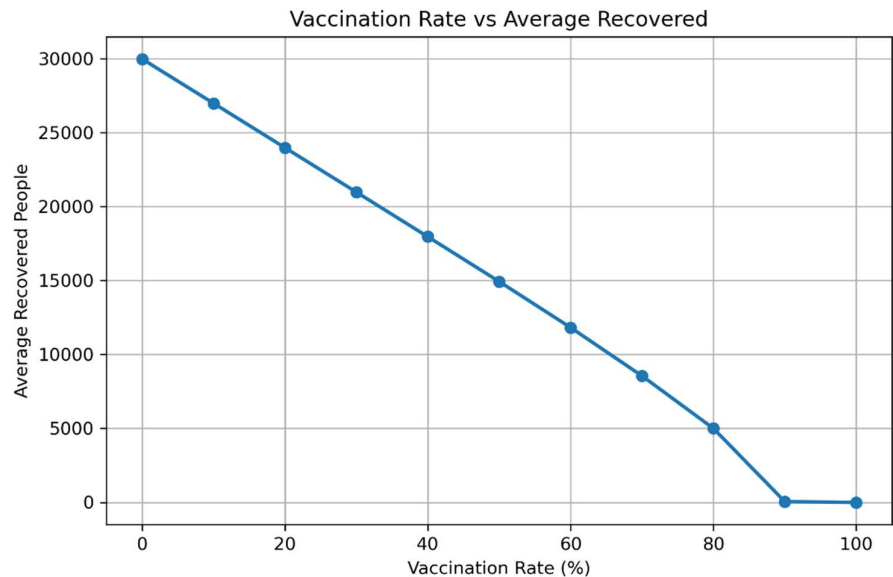


Figure 3: Vaccination rate versus average recovered population

The vaccinated population increases almost linearly with the configured vaccination rate, as shown in Figure 4. For a population of 30,000 persons, every 10 percentage-point increase corresponds to approximately 3,000 additional vaccinated persons. This confirms that the initial vaccination configuration is reflected correctly in the simulated population.

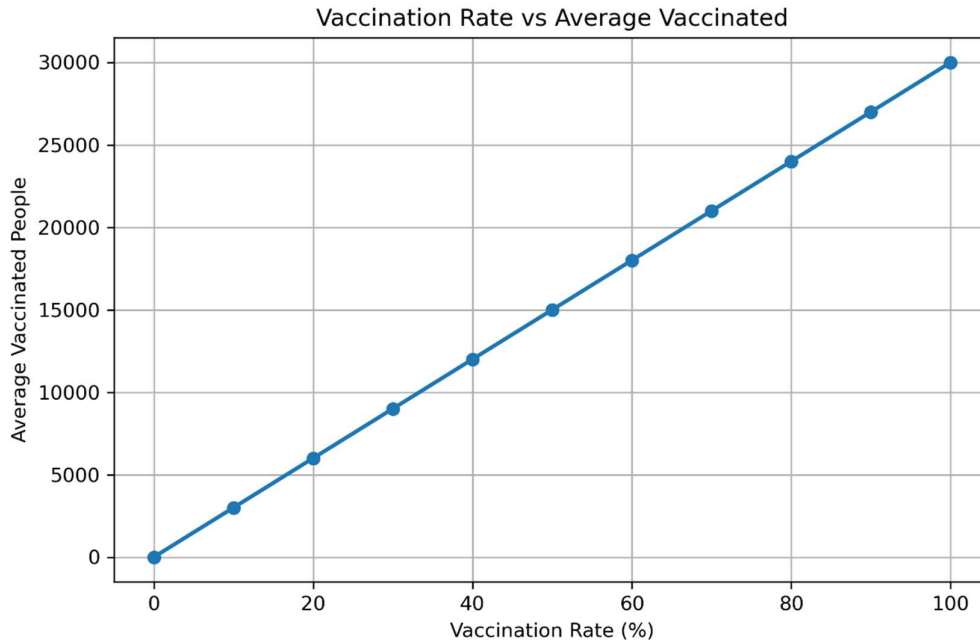


Figure 4: Vaccination rate versus average vaccinated population

4. Discussion:

The results demonstrate a clear herd-immunity effect. At low and moderate vaccination rates, the disease can still propagate through a large susceptible population. As vaccination coverage increases, fewer people become infected and consequently fewer people enter the recovered state. The sharp reduction in recovered persons between 80% and 90% vaccination indicates that the simulated system is approaching a critical suppression region. Assuming six contacts per infectious person per day, the simple estimate $R_n = D \times b \times C$ gives $R_n = 7 \times 0.3 \times 6 = 12.6$. The corresponding idealised herd-immunity threshold is $1 - 1/R_n \approx 92.1\%$. The simulation shows disease spread becoming very small around 90% vaccination, which is broadly consistent with this theoretical estimate. Exact agreement is not expected because the simulation is stochastic and uses discrete person-to-person contacts. The disease-extinction plot also highlights the influence of stochastic dynamics. Close to the threshold, outbreaks may contain only a small number of active infections but can persist for a relatively long time. Random contact selection can cause an infection chain to terminate early or survive for additional timesteps. Therefore, average extinction time should be interpreted together with the recovered-person curve rather than as an independent measure of epidemic severity. The model is intentionally simplified. It assumes uniform random mixing, constant transmissibility, permanent immunity after recovery, and fully effective vaccination. Real populations contain contact networks, behavioural changes and heterogeneous susceptibility.

Nevertheless, the simulation provides a useful computational experiment for studying how vaccination changes epidemic dynamics.

5. Conclusion:

This project investigated herd immunity using an explicit C++ disease simulation with a population of 30,000 persons, a disease duration of 7 days and transmissibility $b = 0.3$. The vaccination sweep showed a strong decline in infections as vaccination coverage increased. The average recovered population approached zero around the 90–100% vaccination range, while the 30% progression curve demonstrated a large epidemic with a pronounced infectious peak. Overall, the numerical results support the expected relationship between vaccination and epidemic suppression. High vaccination coverage substantially reduces the susceptible population and interrupts sustained transmission. The experiment also shows that stochastic extinction time can behave non-linearly near the threshold, emphasising the value of analysing several simulation metrics together.